nature human behaviour

ARTICLES

https://doi.org/10.1038/s41562-022-01289-8

Check for updates

OPEN

Equitable access to COVID-19 vaccines makes a life-saving difference to all countries

Yang Ye[1], Qingpeng Zhang[1] ✉, Xuan Wei[2], Zhidong Cao[3,4], Hsiang-Yu Yuan[5,6] and Daniel Dajun Zeng[3,4] ✉

Despite broad agreement on the negative consequences of vaccine inequity, the distribution of COVID-19 vaccines is imbalanced. Access to vaccines in high-income countries (HICs) is far greater than in low- and middle-income countries (LMICs). As a result, there continue to be high rates of COVID-19 infections and deaths in LMICs. In addition, recent mutant COVID-19 outbreaks may counteract advances in epidemic control and economic recovery in HICs. To explore the consequences of vaccine (in)equity in the face of evolving COVID-19 strains, we examine vaccine allocation strategies using a multistrain metapopulation model. Our results show that vaccine inequity provides only limited and short-term benefits to HICs. Sharper disparities in vaccine allocation between HICs and LMICs lead to earlier and larger outbreaks of new waves. Equitable vaccine allocation strategies, in contrast, substantially curb the spread of new strains. For HICs, making immediate and generous vaccine donations to LMICs is a practical pathway to protect everyone.

Coronavirus disease 2019 (COVID-19) has had devastating impacts on economies, education, health care and social activities in an increasingly globalized world[1,2]. Non-pharmaceutical public health measures, such as border restrictions, social distancing and school closures, have been shown to be insufficient to fully contain the pandemic. Vaccines for COVID-19 are much needed to stop the spread and mutation of the disease and to reopen the world. Because of globalization, there will always be the risk of new outbreaks if vaccines are not widely available. As of 31 December 2021, 31 different vaccines had been approved by at least one country (https://covid19.trackvaccines.org/). More than 331 vaccine candidates are in development, of which 137 are in the clinical development phase[3].

COVID-19 vaccination campaigns were underway worldwide in 2021. As of 31 December 2021, more than nine billion COVID-19 vaccination doses had been administered worldwide (approximately 116 doses per 100 people)[4]. However, country-level and regional vaccination rates are unbalanced. Over 70% of people in high-income countries (HICs) are fully vaccinated against COVID-19; in low-income countries, that number is 4%. Although no country has reported a fully vaccinated population, HICs seem to have access to enough vaccines to vaccinate their populations several times over, leaving many low- and middle-income countries (LMICs) struggling to obtain vaccine supplies to vaccinate their population even once (https://launchandscalefaster.org/COVID-19).

Many researchers and public health experts have warned of the negative consequences of global vaccine inequity[5–10]. Pandemics know no borders, and the public health and economic costs of inequitable vaccine allocation will be borne by all countries in the end. It has been shown in the context of influenza that cross-border vaccination subsidies could provide substantial indirect protection to countries donating vaccines[11,12]. Offering influenza vaccines to neighbouring countries can substantially reduce infections and deaths in both donating and receiving countries. Although many HICs have participated in the COVAX Facility (a joint fund backed by the World Health Organization to ensure equitable distribution of COVID-19 vaccines[13,14]), there remains an acute imbalance in the distribution of COVID-19 vaccines worldwide. Global vaccine inequity is twofold: first, due to the voluntary nature of COVAX, HICs continue to prioritize bilateral deals with vaccine suppliers, leaving scarce vaccine supplies for COVAX[15,16]; second, HICs seem to have underestimated the threat of new strains[17–21] and thus are racing to vaccinate their entire populations and expand booster-shot programmes[22–25] rather than donate vaccines to LMICs to suppress the emergence of new strains. The emergence of the Delta strain[26,27] and the Omicron strain[21] highlights the importance of (1) obtaining a better data-driven understanding of the role of global vaccine equity in preventing the emergence and spread of new strains and (2) identifying a practical pathway to global vaccine equity that reduces morbidity and mortality in both HICs and LMICs. The objective is to transform the global vaccine distribution from a "zero-sum game"[9] to a "cooperative game"[28]. Such a game-theoretic approach has been applied to the control of other pathogens[29–34]. Numerical and analytical results based on hypothetical networks show that the optimal drug/vaccine coordination can reduce epidemic size and overall financial burden of infection for all countries. However, data-driven research on global vaccine coordination in real-world human mobility networks is rare, particularly in the context of the COVID-19 pandemic with viral mutations.

To address these challenges, we propose a multistrain metapopulation model to examine how the pandemic trajectory unfolds under different global vaccine allocation strategies. Harnessing real-world air traffic data, we construct a global mobility network to model human movement across countries. To model the emergence of new

[1]School of Data Science, City University of Hong Kong, Hong Kong, China. [2]Antai College of Economics and Management, Shanghai Jiao Tong University, Shanghai, China. [3]The State Key Laboratory of Management and Control for Complex Systems, Institute of Automation, Chinese Academy of Sciences, Beijing, China. [4]School of Artificial Intelligence, University of Chinese Academy of Sciences, Beijing, China. [5]Department of Biomedical Sciences, Jockey Club College of Veterinary Medicine and Life Sciences, City University of Hong Kong, Hong Kong, China. [6]Centre for Applied One Health Research and Policy Advice, Jockey Club College of Veterinary Medicine and Life Sciences, City University of Hong Kong, Hong Kong, China. ✉e-mail: qingpeng.zhang@cityu.edu.hk; dajun.zeng@ia.ac.cn